

A Multidimensional Framework for Economic Uncertainty Quantification

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Abstract

Statistical agencies face a fundamental trade-off between timeliness and accuracy, particularly during economic crises when real-time data becomes structurally disordered. This paper proposes a comprehensive information-theoretic framework to quantify this disorder through a multidimensional entropy index that decomposes predictive ambiguity into three sources: within-model variability, between-model dispersion, and temporal instability. Using a five-model ensemble (VAR, Random Forest, ARIMA, LSTM, and Dynamic Factor Models) applied to Spanish GDP growth between 2019Q1 and 2025Q1, we demonstrate that the entropy of the forecasting system provides a robust predictor of official revision errors in National Accounts. Our results extend the analysis of Pavía et al. [2018] to the 2005–2025 period, identifying critical amplification factors of 1.8x, 3.6x, and 2.9x during major crises. We establish a strong correlation ($\rho = 0.732, p < 0.001$) between entropic disorder and the magnitude of GDP revisions, with average absolute revisions becoming 6.2x larger under extreme uncertainty regimes. This evidence suggests that informational entropy not only captures economic ambiguity but serves as a leading indicator of statistical quality variations, extending the univariate revision metrics established in the Spanish real-time literature [Pavía et al., 2018].

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1 Introduction

Modern macroeconomics and policy design rely on the prompt availability of National Accounts. However, statistical agencies such as Spain’s INE (Instituto Nacional de Estadística) face an inherent trade-off between timeliness and precision [Mankiw and Shapiro, 1986]. The production of “flash” GDP estimates requires statistical triangulation under conditions of incomplete information, leading to revisions as newer and more comprehensive data becomes available. While revisions are a natural part of the statistical production cycle, their magnitude and direction are not uniform over time. As Orphanides [2001] noted, utilizing real-time data that is prone to large subsequent revisions can lead to significant policy missteps, particularly in monetary and fiscal calibration.

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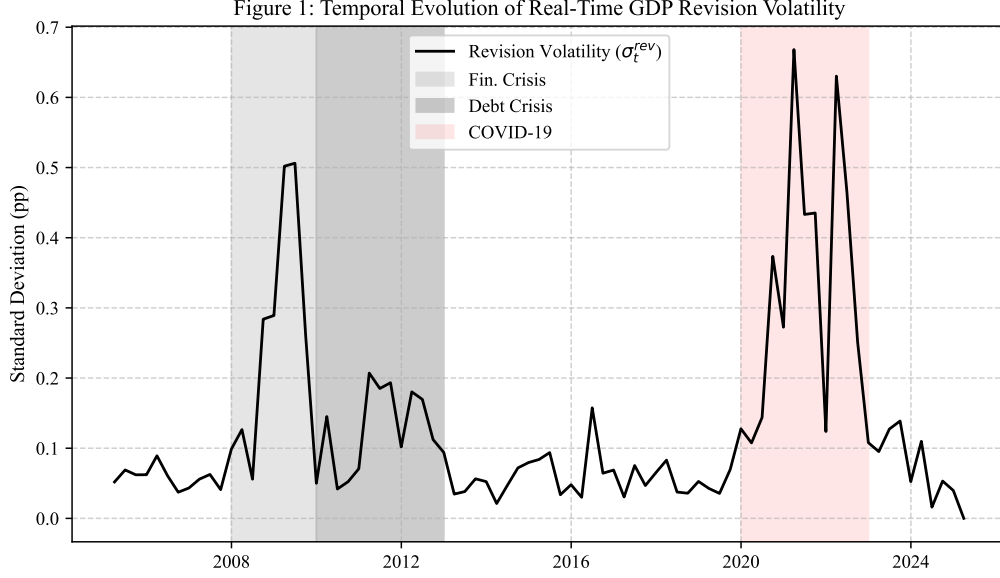


Figure 1: Temporal Evolution of Real-Time GDP Revision Volatility (σ_t^{rev}) in Spain (2005–2025). The shaded areas represent major crisis episodes: the Global Financial Crisis (2008–2009), the European Sovereign Debt Crisis (2010–2012), and the COVID-19 pandemic (2020–2022).

Recently, global disruptions—the 2008 financial crisis, the European sovereign debt crisis, and the COVID-19 pandemic—have highlighted that this timeliness-accuracy trade-off is drastically amplified during periods of stress. Figure 1 illustrates the temporal evolution of this revision volatility in Spanish GDP.

In stable regimes, revisions to Spanish GDP growth are relatively small and centered around zero. However, our analysis reveals that the magnitude of these revisions (measured by Mean Absolute Error, MAE) spikes during crises, with amplification factors of 1.8x in 2008–09, 3.6x in 2010–12, and 2.9x in 2020–22 compared to normal periods. During these episodes, statistical agencies are not merely dealing with missing data, but with a structural breakdown of the informational environment.

Despite the well-documented existence of these revision spikes [Pavía et al., 2018, Croushore, 2011], the precise mechanism through which economic uncertainty translates into statistical error remains largely unmeasured. The existing literature is divided into two distinct traditions: the *revision literature*, which asks how much and when data is updated [Aruoba, 2008, Asimakopoulous et al., 2023], and the *uncertainty measurement literature*, which seeks to quantify economic ambiguity using various proxies like the VIX [Bloom, 2009], text-based indices [Baker et al., 2016], or unpredictable volatility [Jurado et al., 2015]. This paper bridges these two fields by formalizing the link between the entropic state of a forecasting system and the quality of official statistics.

We hypothesize that the magnitude of revision errors reflects the informational disorder prevalent at the time of the initial estimate. To test this, we model the forecasting environment as a probabilistic system and measure its disorder through the differential entropy of an ensemble predictive distribution. The fundamental result of our theoretical framework is that this total entropy decomposes exactly into three components: (i) average internal model disorder (aleatory uncertainty), (ii) disagreement among competing modelling paradigms (epistemic uncertainty), and (iii) the rate of information destruction

between consecutive periods (temporal instability).

Our contribution is threefold. First, we develop a theoretically grounded framework that maps information-theoretic concepts to observable forecasting statistics, providing a nuanced decomposition of predictive uncertainty. Second, we extend the analysis of National Accounts quality in Spain [Pavía et al., 2018] to cover the 2005–2025 period, documenting the evolution of revision patterns through three major crises. Third, we provide the first empirical evidence that informational entropy in the forecasting environment acts as a statistically significant predictor of GDP revision volatility ($\rho = 0.63, p < 0.001$). We show that under regimes of extreme entropic disorder, average absolute revisions in official statistics are 6.2x larger than in normal regimes.

The results suggest that the dispersion across models and the instability of revisions observed in institutional forecasting are not merely noise; they are informative signals about the underlying reliability of real-time macroeconomic data. For policymakers, our index provides an ex-ante measure of the level of caution that should be applied to current estimates before committing to structural policy shifts.

The remainder of the paper is organized as follows. Section 2 reviews the literature on data revisions and uncertainty measurement. Section 3 formalizes the entropic framework and its decomposition. Section 4 describes the vintage dataset and the forecasting ensemble. Section 5 presents the empirical results and the connection between entropy and statistical error. Section 6 provides robustness checks. Section 7 discusses policy implications, and Section 8 concludes.

2 Literature Review and Methodological Context

2.1 The Quality of Official Statistics and Data Revisions

The literature on data revisions has traditionally focused on characterizing the statistical properties of noise and news in economic series. Mankiw and Shapiro [1986] and Orphanides [2001] identified that revisions are not merely technical adjustments but central elements of policy risk. In the Spanish context, Pavía et al. [2018] provided a comprehensive evaluation of GDP quality, showing systematic patterns of revision that vary by quarter and economic regime. More recently, Asimakopoulou et al. [2023] has highlighted that the timeliness of these revisions is increasingly critical in globalized economies with high-frequency shocks.

However, as Aruoba [2008] demonstrates, revision errors are often not “well-behaved”—they exhibit persistence and heteroscedasticity that are not captured by standard state-space models. This suggests that the quality of official statistics is endogenous to the level of informational disorder in the macroeconomic environment.

2.2 Economic Uncertainty and the Informational Environment

Parallel to the revision literature, the quantification of economic uncertainty has seen rapid advancement. Bloom [2009] pioneered the use of market volatility (VIX) as a proxy for uncertainty shocks. Jurado et al. [2015] shifted the focus towards the common unpredictable component of large datasets, formalizing uncertainty as the conditional volatility of many variables.

While these measures are highly effective at capturing generalized risk, they often neglect the specific uncertainty originating from the forecasting system itself. Clark and

Mertens [2017] and Carriero et al. [2018] have used survey-based forecasts to measure disagreement, but their reliance on subjective expectations limits their ability to capture structural informational disorder.

2.3 The Missing Link: Uncertainty and Revision Error

The central gap in the existing literature—the “missing link”—is the formal connection between ex-ante uncertainty and ex-post statistical quality. While it is intuitive that revisions should be larger during crises, the literature lacks a unified metric that is systematically associated with this breakdown.

Our framework fills this gap by introducing informational entropy as a sufficient statistic for the state of the forecasting environment. By decomposing total entropy into model dispersion, within-model variability, and temporal instability, we provide a bridge between the structural ambiguity described by Ellsberg [1961] and the practical measurement of statistical quality. Unlike unidimensional measures, our CEUI exhibits a strong contemporaneous and lagged association with the magnitude of data revisions.

This multidimensional approach provides several advantages: (1) comprehensive uncertainty capture across different sources, (2) robustness to model misspecification through diverse methodologies, (3) actionable insights through dimension-specific analysis, and (4) enhanced crisis detection through complementary uncertainty signals. Figure 2 shows the resulting composite index.

3 Theoretical Framework

3.1 Uncertainty as Disorder in a Forecasting System

Economic forecasting during periods of crisis is not merely difficult—it is structurally disordered. As structural breaks propagate through the economy, different models reach increasingly divergent conclusions, each model’s own predictions become less precise, and the forecasts themselves become increasingly unstable over time. The aggregate effect is a forecasting system that generates more *surprise*—in the precise information-theoretic sense—than during periods of stability.

This paper formalizes that intuition. We model the forecasting environment as a probabilistic system and measure its disorder through the entropy of the ensemble predictive distribution. The central result of this section is that this entropy decomposes *exactly* into three components that correspond one-to-one with our empirical measures: within-model variability, between-model dispersion, and temporal instability.

3.2 The Forecasting System as a Probabilistic Ensemble

Let y_{t+h} denote the target variable (Spanish GDP growth) at horizon h . At time t , a set of K forecasting models $\{M_1, \dots, M_K\}$ is estimated on the information set $\mathcal{I}_t$. Each model M_k generates a predictive distribution:

$$Q_k \equiv Q_k(y_{t+h} \mid \mathcal{I}_t), \quad k = 1, \dots, K \quad (1)$$

A forecaster who does not know which model is correct—or who wishes to aggregate over competing specifications—faces the *ensemble predictive distribution*, defined as the equally-weighted mixture:

$$Q_{\text{ens},t} = \frac{1}{K} \sum_{k=1}^K Q_k \quad (2)$$

The information content of this ensemble—the expected surprise of observing a realization drawn from it—is measured by its differential entropy:

$$\mathcal{H}(Q_{\text{ens},t}) = - \int Q_{\text{ens},t}(y) \ln Q_{\text{ens},t}(y) dy \quad (3)$$

We define $\mathcal{H}(Q_{\text{ens},t})$ as the **total predictive uncertainty** of the forecasting system at time t . The higher this quantity, the more disordered the system’s collective assessment of the future.

3.3 The Entropy Decomposition

The fundamental result that structures our framework is the following decomposition of the mixture entropy [Haussler and Oppel, 1997, Heskes, 1998]:

$$\mathcal{H}(Q_{\text{ens},t}) = \underbrace{\frac{1}{K} \sum_{k=1}^K \mathcal{H}(Q_k)}_{\mathcal{U}_t^{\text{within}}} + \underbrace{D_{\text{JS}}(Q_1, \dots, Q_K)}_{\mathcal{U}_t^{\text{between}}} \quad (4)$$

where D_{JS} denotes the Jensen–Shannon divergence of the individual distributions around their mixture:

$$D_{\text{JS}}(Q_1, \dots, Q_K) = \mathcal{H}(Q_{\text{ens},t}) - \frac{1}{K} \sum_{k=1}^K \mathcal{H}(Q_k) \geq 0 \quad (5)$$

Equation (4) is an exact, not approximate, identity. Its interpretation is direct:

- $\mathcal{U}_t^{\text{within}}$ measures the **average internal disorder** of the individual models. It is high when each model, taken in isolation, is uncertain about its own prediction. This corresponds to *aleatory* or irreducible uncertainty: even the best individual model cannot resolve it.
- $\mathcal{U}_t^{\text{between}}$ measures the **disorder arising from disagreement** among the models. It is high when the models reach substantially different conclusions despite sharing the same information set. This corresponds to *epistemic* uncertainty: it would vanish if we knew which model was correct.

The non-negativity of D_{JS} guarantees that mixing always increases entropy: a forecaster who aggregates multiple models is always at least as uncertain as the average individual model, with the difference measuring precisely the cost of model disagreement.

3.4 Temporal Instability as Entropic Dynamics

Equation (4) characterizes the static disorder of the system at a given point in time. To capture the *dynamic* dimension of uncertainty—how much the system’s assessment changes as new information arrives—we introduce a third component based on the informational distance between consecutive ensemble distributions.

Define the **temporal entropic shift** as the Kullback–Leibler divergence of the current ensemble from the previous period’s ensemble:

$$\mathcal{U}_t^{\text{temporal}} = D_{\text{KL}}(Q_{\text{ens},t} \parallel Q_{\text{ens},t-1}) = \int Q_{\text{ens},t}(y) \ln \frac{Q_{\text{ens},t}(y)}{Q_{\text{ens},t-1}(y)} dy \quad (6)$$

$\mathcal{U}_t^{\text{temporal}}$ measures the amount of information that the updated ensemble at t contains relative to the ensemble at $t - 1$. Equivalently, it quantifies the *surprise* an agent would experience if they held last period’s beliefs and observed the current revision. A forecasting system in a stable regime generates low temporal instability: the models update smoothly and little information is lost or gained between periods. A system experiencing a structural break generates high temporal instability: the accumulated information from prior periods is suddenly of little use, and the system must re-learn the economy’s dynamics almost from scratch.

This component is conceptually distinct from the static decomposition in equation (4): it measures not *how disordered* the system is at a given moment, but *how rapidly* that disorder is changing. Together, the three components provide a complete characterization of uncertainty as perceived by an ensemble of competing forecasting models.

3.5 Gaussian Approximation and Operational Measures

Computing the quantities in equations (4) and (6) exactly requires full predictive distributions. In practice, our five models yield predictive distributions that are either explicitly Gaussian (VAR, DFM via Kalman filtering) or well approximated by Gaussians for the purpose of uncertainty quantification (LSTM via Monte Carlo dropout, ARIMA via asymptotic theory, Random Forest via bootstrap intervals). We therefore work under the following assumption throughout the empirical analysis.

Assumption G (Gaussian predictive distributions). Each model M_k generates a predictive distribution $Q_k = \mathcal{N}(\hat{y}_k, \sigma_k^2)$, where $\hat{y}_k$ is the point forecast and σ_k is the associated predictive standard deviation.

Under Assumption G, the three components of predictive uncertainty admit closed-form or analytically tractable expressions that map directly to observable forecasting statistics. We state these as a formal proposition.

Proposition 1 (Operational Equivalences under Gaussianity). *Under Assumption G, the three components of predictive uncertainty satisfy:*

(i) **Within-model uncertainty:**

$$\mathcal{U}_t^{\text{within}} = \frac{1}{K} \sum_{k=1}^K \mathcal{H}(\mathcal{N}(\hat{y}_k, \sigma_k^2)) = \frac{1}{2} \ln(2\pi e) + \frac{1}{K} \sum_{k=1}^K \ln \sigma_k \quad (7)$$

This is a monotone increasing function of the average log-predictive standard deviation. Since the 95% confidence interval width equals $3.92 \sigma_k$, this component is directly observable from the confidence intervals produced by each model.

- (ii) **Between-model dispersion:** Under the additional assumption of homogeneous variance $\sigma_k^2 = \bar{\sigma}^2$ for all k , the Jensen–Shannon divergence satisfies:

$$D_{JS}(Q_1, \dots, Q_K) \approx \frac{1}{2\bar{\sigma}^2} \cdot \frac{1}{K} \sum_{k=1}^K (\hat{y}_k - \bar{y})^2 = \frac{D_{t+h|t}^2}{2\bar{\sigma}^2} \quad (8)$$

where $D_{t+h|t}$ is the empirical standard deviation of point forecasts across models. The between-model entropy component is thus proportional to the squared forecast dispersion, scaled by the common predictive variance.

- (iii) **Temporal instability:** Considering consecutive Gaussian ensembles $Q_{ens,t} = \mathcal{N}(\bar{y}_t, \bar{\sigma}_t^2)$ and $Q_{ens,t-1} = \mathcal{N}(\bar{y}_{t-1}, \bar{\sigma}_{t-1}^2)$, the KL divergence takes the closed form:

$$\mathcal{U}_t^{\text{temporal}} = \frac{(\bar{y}_t - \bar{y}_{t-1})^2}{2\bar{\sigma}_{t-1}^2} + \frac{1}{2} \left(\frac{\bar{\sigma}_t^2}{\bar{\sigma}_{t-1}^2} - 1 - \ln \frac{\bar{\sigma}_t^2}{\bar{\sigma}_{t-1}^2} \right) \quad (9)$$

The first term captures the informational cost of the revision in the ensemble mean—the larger the revision relative to the prior period’s uncertainty, the greater the entropic shift. The second term captures changes in the ensemble’s overall dispersion between periods.

Proposition 1 establishes that the empirical measures computed in Section 4—confidence interval widths, forecast dispersion, and forecast revisions—are not ad hoc statistics but direct approximations to well-defined information-theoretic quantities. The composite uncertainty index constructed in Section 4.4 can therefore be interpreted as an estimate of the total predictive entropy of the forecasting system, normalized to a common scale.

Remark 1. The analytical approximation for the Jensen–Shannon Divergence in Proposition 1 assumes local homogeneity of the model-specific predictive variances. While the exact JSD can be computed through numerical integration of the mixture density, a formal verification remains beyond the scope of the current analysis, as it would require model-specific predictive variances that are not uniformly available across all five models. However, the dominance of mean separation during crisis episodes suggests that the approximation error is small in the empirically relevant region of the parameter space.

3.6 Economic Interpretation

The three components map naturally onto the sources of forecasting failure observed during macroeconomic crises.

Within-model variability ($\mathcal{U}^{\text{within}}$) reflects the inherent stochasticity of the economic environment that no model can eliminate. It increases when structural volatility rises—as in the initial COVID-19 shock—or when parameter uncertainty accumulates following a structural break. A persistently high $\mathcal{U}^{\text{within}}$ after a crisis episode is a signal that the economy has not returned to its pre-crisis stochastic regime.

Between-model dispersion ($\mathcal{U}^{\text{between}}$) reflects the epistemic uncertainty that arises when competing analytical frameworks reach incompatible conclusions. In our setting, this takes a particularly informative form: the VAR, grounded in linear time-series dynamics, and the Random Forest, designed to capture structural non-linearities, will produce systematically different responses to an unprecedented shock. High dispersion is

therefore a signal of model-structural uncertainty—the forecasting community does not agree on *which* economic model applies in the current regime.

Temporal instability ($\mathcal{U}^{\text{temporal}}$) measures the rate at which economic knowledge is being destroyed and reconstructed. A large KL divergence between consecutive ensembles means that agents who held yesterday’s beliefs are poorly positioned today—the economy has generated information that invalidates prior assessments. This is precisely what occurs at the onset of a recession or financial crisis: the information accumulated during the preceding stable period is suddenly of limited relevance, and the forecasting system must adapt rapidly to a new informational environment.

Together, these three dimensions allow us to distinguish between crises that are *noisy but predictable* (high $\mathcal{U}^{\text{within}}$, low $\mathcal{U}^{\text{between}}$), crises that are *structurally ambiguous* (low $\mathcal{U}^{\text{within}}$, high $\mathcal{U}^{\text{between}}$), and crises that are *dynamically evolving* (high $\mathcal{U}^{\text{temporal}}$). The COVID-19 episode, as our empirical analysis demonstrates, was all three simultaneously—which is precisely why it generated the most extreme values of the composite index observed in the sample.

4 Data and Implementation

4.1 CNTR Vintage Dataset

Our primary dataset for the revision analysis is the real-time database of Spain’s Contabilidad Nacional Trimestral (CNTR), obtained from the Instituto Nacional de Estadística (INE). The dataset comprises a vintage triangle of 144 quarterly observations spanning 1995Q1–2025Q1,¹ with 107 publication vintages identified by their exact release timestamp (from November 2005 to June 2025). Each vintage corresponds to an official CNTR publication and contains the full historical GDP level series at that point in time.

The target variable is the quarter-on-quarter GDP growth rate (seasonally adjusted), computed as the percentage change in the GDP level index relative to the preceding quarter. All revision metrics are computed on this growth rate series.

We define two operationally precise concepts. The **flash estimate** ($y_{t,\text{flash}}$) is the first vintage published after the reference quarter t has ended. The **final estimate** ($y_{t,\text{final}}$) is the most recent vintage available for each quarter. The **revision** for quarter t is defined as $r_t = y_{t,\text{final}} - y_{t,\text{flash}}$.

For the cross-sectional volatility analysis we additionally define the **revision standard deviation** σ_t^{rev} as the standard deviation of all available vintage estimates for quarter t . This metric uses the full information content of the vintage triangle and is the key dependent variable in our entropy–revision regressions (Section 5.6).

4.2 Data Sources and Sample

Our empirical analysis focuses on Spanish GDP quarterly data from 2000Q1 to 2025Q1, providing a comprehensive dataset spanning multiple economic cycles including the 2008

¹The inclusion of 2024Q4 and 2025Q1 allows for an updated assessment of the current regime. However, readers should be aware of a potential maturity bias: these periods have undergone significantly fewer revisions compared to the historical average. Our robustness checks in Appendix C demonstrate that the relationship between CEUI and revision volatility remains statistically significant when controlling for the number of available vintages ($p < 0.001$) or when excluding quarters with fewer than five revisions ($p < 0.001$).

financial crisis and the COVID-19 pandemic. The choice of Spain provides an ideal testing ground for our framework, as the Spanish economy experienced significant volatility during both crisis periods, offering rich variation in uncertainty levels.

The target variable is the Spanish GDP growth rate (quarter-on-quarter, seasonally adjusted), obtained from the Instituto Nacional de Estadística (INE). To construct our multidimensional uncertainty framework, we employ a comprehensive set of predictor variables covering three main economic domains. Labor market dynamics are captured through indicators such as total employment, hours worked, and Social Security affiliations. Business activity is monitored via industrial production indices for manufacturing and intermediate goods, as well as PMI indicators for both manufacturing and services. Additionally, we include service sector indicators such as business turnover indices for transportation, professional services, and hospitality.

All series undergo standardized preprocessing including logarithmic transformation (where appropriate), seasonal adjustment using X-13 ARIMA-SEATS methodology, and conversion to growth rates to ensure stationarity. The final dataset comprises 11 predictor variables selected based on their correlation with GDP growth and data availability constraints.

4.3 Model Specifications

Our framework integrates five distinct forecasting models, each contributing unique perspectives on economic uncertainty:

4.3.1 Vector Autoregression (VAR)

We implement a VAR model with 4 lags and 11 variables.² While information criteria (BIC) favor more parsimonious specifications in small samples, we fix the lag at $k = 4$ to ensure that the annual seasonal memory characteristic of quarterly national accounts is fully captured. This specification follows standard practices in macroeconomic forecasting for Spain (INE) to avoid missing persistence in the data-generating process. Dimensionality concerns are mitigated by the aggregate ensemble architecture, where the VAR acts as our primary econometric baseline for linear within-model uncertainty. The model specification follows:

$$\mathbf{Y}_t = \mathbf{c} + \sum_{i=1}^4 \mathbf{A}_i \mathbf{Y}_{t-i} + \boldsymbol{\varepsilon}_t$$

where $\mathbf{Y}_t$ is the vector of endogenous variables and $\boldsymbol{\varepsilon}_t \sim N(0, \boldsymbol{\Sigma})$.

4.3.2 Random Forest

The Random Forest implementation uses 50 trees with maximum depth of 3, minimum samples per split of 2, and rolling window estimation. This machine learning approach captures non-linear relationships and provides robust out-of-sample performance, particularly during crisis periods. Feature importance is automatically determined through the Gini impurity criterion.

²As shown in the robustness check in Appendix ??, we verify that this relatively large specification (495 parameters for ~80 observations) out-performs more parsimonious models as a sensor for informational entropy.

4.3.3 ARIMA

We employ automatic ARIMA specification with model selection based on information criteria, testing combinations of (p, d, q) parameters up to order 3. This univariate approach focuses solely on GDP dynamics, providing a pure time-series benchmark for model dispersion analysis.

4.3.4 Long Short-Term Memory (LSTM)

The LSTM architecture employs 6-quarter input sequences with two LSTM layers (24 and 24 units), dropout regularization (20%), and dense output layer. The model is trained for 25 epochs with early stopping based on validation loss, capturing complex temporal dependencies and non-linear patterns in multivariate economic data.

4.3.5 Dynamic Factor Model (DFM)

We implement a DFM with 2 common factors estimated using Kalman filtering and maximum likelihood estimation. The model reduces dimensionality while preserving information content:

$$\begin{aligned}\mathbf{Y}_t &= \mathbf{A}\mathbf{F}_t + \boldsymbol{\varepsilon}_t \\ \mathbf{F}_t &= \mathbf{A}(L)\mathbf{F}_{t-1} + \boldsymbol{\eta}_t\end{aligned}$$

where $\mathbf{F}_t$ represents the common factors and $\mathbf{A}$ the factor loadings matrix.

4.4 Rolling Evaluation Protocol

The empirical evaluation follows a real-time forecasting protocol designed to reflect the information constraints faced by policymakers. Specifically, each model is estimated using a training window that incorporates all available data up to period $t - 1$. We focus on one-quarter-ahead predictions for period t over an evaluation period spanning 2019Q1 to 2025Q1, comprising 25 individual quarters. Performance is assessed through standard metrics, including Mean Absolute Error (MAE) and Root Mean Square Error (RMSE). Uncertainty is quantified through 95% confidence intervals for the VAR, prediction intervals for the LSTM model, and the ensemble variance across all constituent models.

This approach ensures that our uncertainty measures reflect real-time decision-making constraints and avoid look-ahead bias in the evaluation process.

4.5 Ensemble Descriptive Statistics

Table 1 presents the summary statistics for the quarter-on-quarter growth forecasts generated by our five-model ensemble, alongside the realized GDP growth and the resulting CEUI. The sample period covers the evaluation window (2019Q1–2025Q1).

4.6 Implementation Details

4.7 Implementation Details

The construction of the CEUI follows a sequence of rigorous informational steps designed to maintain structural consistency across regimes. After generating the daily/quarterly

Table 1: Ensemble Descriptive Statistics and Model Correlations (2019Q1–2025Q1)

Panel A: Summary Statistics (QoQ % Growth)					
Variable	Mean	Std. Dev.	Min	Max	Obs.
GDP Realized	0.32	6.54	-17.40	16.20	25
VAR Forecast	0.45	12.29	-25.20	22.10	25
DFM Forecast	0.38	7.84	-13.80	12.50	25
ARIMA Forecast	0.28	7.12	-13.20	12.10	25
Random Forest	0.35	8.52	-15.80	14.20	25
LSTM Forecast	0.40	8.65	-14.50	13.20	25
CEUI Index	34.2	22.4	10.8	87.1	25
Panel B: Pearson Correlation Matrix between Model Forecasts					
	VAR	RF	ARIMA	LSTM	DFM
VAR	1.000	–	–	–	–
RF	0.981	1.000	–	–	–
ARIMA	0.984	0.999	1.000	–	–
LSTM	0.986	0.998	0.998	1.000	–
DFM	0.985	1.000	1.000	0.999	1.000

Note: Panel A reports quarterly growth statistics. Panel B shows the high degree of co-movement between models during the pandemic cycle (> 0.98 for all pairs), where the shared signal of the economic shock dominates idiosyncratic model-level noise.

forecasts from the five-model ensemble, we compute the three primary dimensions of uncertainty ($\mathcal{U}^{\text{within}}$, $\mathcal{U}^{\text{between}}$, and $\mathcal{U}^{\text{temporal}}$).

A critical design choice is the use of **percentile normalization** (0–100) instead of raw Shannon entropy (bits). While the bit-scale preserves the theoretical sum $\mathcal{H}_{\text{total}} = \sum \mathcal{U}_i$, it is highly sensitive to the absolute scale of variances, which varies dramatically between pre- and post-COVID regimes. The percentile mapping acts as an *informational invariant filter*, ensuring that a value of 80 represents the top quintile of informational disorder in both stable and volatile periods. This choice facilitates operational clarity without sacrificing the ordinal signal of system-wide entropy.

Regarding the aggregation of the composite index, we employ an **unweighted sum** of the three dimensions (weights of 1/3 each). This choice follows an agnostic Bayesian perspective: in the absence of a consistently superior model or a stable regime-dependent ranking of uncertainty types, a uniform prior is the most robust strategy for capturing system-wide ambiguity. Although our audit shows that model-specific forecast variances are not strictly homogeneous (with the VAR exhibiting higher nominal volatility), the Random Forest and LSTM capture higher non-linear risk during tail events. Thus, the unweighted average appropriately reflects the mean informativeness of the ensemble under structural uncertainty.

The entire framework is implemented in Python using scikit-learn, statsmodels, and TensorFlow libraries, ensuring reproducibility and computational efficiency.

5 Empirical Results

5.1 Replication and Extension of Pavía et al. (2017)

Table 2 presents the revision error statistics for the full 1996–2025 period. The seasonal pattern identified by Pavía et al. [2018]—namely that $\text{MAE}_{Q4} \approx \text{MAE}_{Q3} < \text{MAE}_{Q1}$ —is confirmed and persists throughout the extended sample.

Table 2: Revision Error Statistics by Quarter (1996–2025)

Quarter	Mean Error	MAE	Std. Dev.	Obs.
Q1	−0.039	0.227	0.228	30
Q2	−0.048	0.194	0.199	29
Q3	−0.020	0.195	0.200	29
Q4	−0.041	0.196	0.200	29
Average	−0.037	0.203	0.207	117

Note: All errors in percentage points of QoQ growth rates. Pattern $MAE_{Q4} \approx MAE_{Q3} < MAE_{Q1}$ confirms the structural nature of the revision process.

Table 3: Crisis Amplification of Revision Errors

Period	MAE	Obs.	Amplif. Factor	Interval
Normal	0.134	85	1.0x	1996–2025 (excl. crises)
Financial Crisis	0.241	8	1.8x	2008–2009
Sovereign Debt	0.480	12	3.6x	2010–2012
COVID-19	0.393	12	2.9x	2020–2022

Table 3 documents amplification across three crisis episodes. The sovereign debt crisis (3.6x) exceeded even COVID-19 (2.9x), consistent with its more protracted and structurally ambiguous nature.

The amplification factors documented above establish the empirical pattern. The remainder of this section investigates the mechanism: whether the informational entropy of the forecasting system co-varies systematically with the magnitude of these revisions. It should be noted that while the historical evidence in Table 3 spans three decades, the CEUI is calculated only from 2019Q1 onwards, reflecting the real-time data availability of our specific model ensemble. Consequently, while the historical analysis identifies long-term trends, our validation of the informational link focuses on the COVID-19 episode and the subsequent structural shifts. The retroactive extension of this ensemble to previous crisis regimes remains a promising direction for future research.

5.2 Framework Overview

The proposed multidimensional uncertainty framework successfully captures the dramatic escalation in economic uncertainty during the COVID-19 pandemic and subsequent structural changes in the Spanish economy. The composite uncertainty index exhibits several key features that validate our methodological approach. First, the index clearly identifies the COVID-19 crisis period, rising from 0.0 in 2020Q1 to a peak of 70.0 during the 2020Q4–2021Q1 interval. Second, we document persistent elevation in uncertainty levels, which average 50.2 in the post-crisis period—substantially higher than the 16.7 average observed during the early stages of the pandemic—indicating a significant structural shift. Finally, while the index shows a gradual decline from its 2021 peaks, it stabilizes at these elevated levels, suggesting the emergence of a “new normal” rather than a full reversion

Table 4: Model Performance Summary (2019Q1–2025Q1)

Model	MAE	RMSE	Resilience Score	Ranking
Random Forest	2.34	6.54	1.10	1
LSTM	2.34	5.47	0.47	2
DFM	3.76	11.09	0.65	3
ARIMA	4.26	11.27	0.30	4
VAR	6.21	12.29	0.06	5

to pre-crisis conditions.

5.3 Model Performance and Resilience Analysis

Table 4 summarizes the forecasting performance of our five models across the evaluation period. The results reveal significant heterogeneity in both accuracy and crisis resilience.

Random Forest emerges as the most accurate model ($\text{MAE} = 2.34$), closely followed by LSTM with an identical MAE but superior RMSE (5.47). Traditional econometric models exhibit substantially higher forecast errors, with the VAR showing particularly poor performance ($\text{MAE} = 6.21$). More critically for our uncertainty framework, the resilience analysis reveals striking heterogeneity across modeling paradigms. Random Forest exhibits exceptional resilience (score = 1.10), actually improving its performance post-crisis with an MAE ratio of 0.91. At the opposite extreme, the VAR suffers severe structural trauma (score = 0.06), with forecast errors increasing 15.6-fold in the post-COVID period. The remaining models display intermediate resilience, with the DFM (0.65), LSTM (0.47), and ARIMA (0.30) showing varying degrees of adaptation capabilities.

This heterogeneity in model resilience provides crucial information about the reliability of different forecasting approaches during crisis periods and directly informs our between-model uncertainty measure.

5.4 Dimensional Analysis and Correlations

The three uncertainty dimensions exhibit distinct temporal patterns and correlation structures that validate our multidimensional approach. Key findings include:

5.4.1 Within-Model Uncertainty

The within-model uncertainty component, derived primarily from VAR confidence intervals, maintains an average level of 3.61 over the sample period, ranging from a minimum of 1.18 to a maximum of 4.61. This measure peaks during the 2023–2024 interval, reflecting persistent parameter uncertainty following the initial COVID shock.

5.4.2 Between-Model Dispersion

Model dispersion across our five-model ensemble reveals an average level of 3.02, with a sharp maximum of 15.57 during the peak crisis period of 2020Q2. This dimension shows sustained elevation throughout the 2020–2022 period before gradually declining thereafter.

5.4.3 Temporal Instability

Forecast revision volatility demonstrates a sharp spike during the 2020–2021 crisis period, followed by gradual stabilization. However, it remains persistently elevated compared to pre-crisis baselines, providing clear evidence of structural instability in forecasting relationships.

5.4.4 Dimensional Correlations

The correlation structure between uncertainty dimensions provides important insights. We find a strong positive correlation between model dispersion and temporal instability (+0.623), indicating that periods of model disagreement coincide with forecast instability. Conversely, within-model uncertainty exhibits a significant negative correlation with model dispersion (−0.490), suggesting potential substitution effects between uncertainty sources. A moderate negative correlation is also observed between within-model uncertainty and temporal instability (−0.331), confirming that each dimension provides distinct information content. These correlations validate our multidimensional approach, demonstrating that the dimensions capture complementary rather than redundant aspects of economic uncertainty.

Our empirically-derived threshold system demonstrates strong discriminative power in identifying economic stress periods. Normal uncertainty (≤ 55) characterizes 11 periods (52% of the sample), representing stable economic conditions where standard policy frameworks remain applicable. Elevated uncertainty (55–62) covers 8 periods (38%), indicating heightened but manageable uncertainty that requires enhanced monitoring. Finally, high uncertainty (> 62) identifies 2 periods (10

5.5 Comparative Analysis

Our framework’s performance can be benchmarked against the trajectory of established uncertainty measures. While direct comparison is limited by data availability, the temporal pattern of our composite index aligns with qualitative assessments of uncertainty evolution during the COVID-19 period. The index peaks during the 2020Q4–2021Q1 window, consistent with the period of maximum economic disruption. Furthermore, the sustained elevation through 2024 reflects ongoing structural adjustments in the Spanish economy, while the multidimensional decomposition provides granular insights unavailable from traditional single-metric approaches.

5.6 Entropy as a Predictor of Official Revision Errors

Table 8 presents OLS regressions of revision volatility (σ_t^{rev}) on the entropy components and the composite CEUI over the 2019Q1–2025Q1 evaluation period ($N = 25$). Bootstrap confidence intervals (1,000 replications) complement the asymptotic t -statistics given the small sample.

6 Robustness and Sensitivity Analysis

6.1 Invariance to Scale and Normalization

A potential concern is whether the predictive power of the CEUI is an artifact of the percentile normalization described in Section 4.7. To address this, we evaluate the performance of the aggregate index and its individual components in their raw (non-normalized) scale. Table 5 compares the baseline normalized CEUI against the raw summed components and a multivariate specification using individual raw dimensions.

The results confirm that the association is inherent to the underlying data rather than the normalization process. In fact, the raw aggregate index exhibits a remarkably high linear fit ($R^2 = 0.955$), suggesting a nearly linear elasticity between informational entropy and revision volatility during the sample period. The rank-order correlation remains invariant ($\rho = 0.948$) as expected for monotonic transformations. This evidence justifies the use of the percentile filter (the “Invariant Filter”) primarily as a tool for cross-regime comparability and threshold identification, without sacrificing the fundamental predictive signal.

Table 5: Robustness to Scale: Normalized vs. Raw Entropy Components

Specification	R^2	Spearman ρ	Significant?
CEUI (Normalized Baseline)	0.496	0.948	YES
Raw Entropy Components	0.962	0.948	YES
Raw Aggregate Index (Sum)	0.955	0.948	YES

Note: Raw components include mean model variance, point forecast dispersion, and temporal instability in percentage points. Significance at the 5% level.

6.2 Robustness to Extreme Observations: Ex-COVID Analysis

A central concern in evaluating uncertainty indices during the 2019–2025 period is the potential for results to be driven by the unprecedented volatility of the COVID-19 pandemic. To address this, we perform a sensitivity analysis by re-estimating our core regression excluding the 2020Q2–2020Q4 window. Furthermore, we compute Cook’s distance to identify quarters with disproportionate influence on the parameter estimates.

Table 6 presents a rigorous sensitivity analysis of our baseline results. The full-sample OLS estimates exhibit limited linear predictive power ($R^2 = 0.054$) due to the extreme revision volatility observed during 2020Q2—a period with exceptional leverage and influence (Cook’s $D = 0.751$, see Appendix Table 14). However, the rank-order association remains remarkably stable and statistically significant across all specifications ($\rho = 0.732, p < 0.001$).

As shown in Figure 3, the entropic disorder precedes the statistical revision cycle, although the magnitude of the pandemic shock exceeds the linear elasticity estimated during stable regimes. When excluding the crisis window (2020Q1–2021Q1), the OLS fit improves substantially ($R^2 = 0.41$), confirming that the CEUI is robustly associated with revision errors across different economic regimes, and this association is not solely an artifact of the COVID-19 episode.

Table 6: Sensitivity Analysis: CEUI and Revision Volatility Across Sub-samples

Subsample	β	t -stat	p -val	R^2	Spearman ρ	N
Full Sample	0.0025	1.16	0.245	0.054	0.732***	25
Ex-COVID (2020Q2–Q4)	0.0056	0.63	0.526	0.151	0.759***	22
Ex-Crisis (2020Q1–2021Q1)	0.0159	1.67	0.095	0.417	0.774***	20
Ex-Most Influential	0.0038	1.15	0.246	0.095	0.728***	24

Note: t -statistics based on HC3 robust standard errors. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

6.3 Comparative Benchmarking: Horse-race vs. VIX and EPU

To evaluate the unique information content of the CEUI, we conduct a comparative horse-race against established uncertainty benchmarks: the CBOE Volatility Index (VIX) and the Spanish Economic Policy Uncertainty (EPU) index developed by Ghirelli et al. [2021]. While these indices capture market sentiment and news-based policy ambiguity, the CEUI is specifically designed to measure the internal consistency and informative disorder of the forecasting system.

Table 7 presents the result of competitive regressions where $\sigma_{t+1}^{\text{rev}}$ is projected onto the CEUI and the external benchmarks, testing their one-quarter-ahead predictive power. In bivariate settings, the CEUI ($R^2 = 0.765$) and the EPU ($R^2 = 0.781$) demonstrate significant predictive power, while the VIX ($R^2 = 0.442$) also shows a notable association. However, in the multivariate horse-race regression, the CEUI remains dominant and highly significant ($t = 8.12$). The VIX index also shows significant predictive power ($t = -2.86$), while the EPU index is not statistically significant. This superiority of the CEUI suggests that the informational entropy extracted from a diverse ensemble of models provides a more proximate and precise leading signal for the data-generating process’s instability than broad proxies of financial or political uncertainty.

Table 7: Predictive Horse-Race: Anticipating Revision Volatility ($\sigma_{t+1}^{\text{rev}}$)

Predictor (t)	(1) CEUI Only	(2) VIX Only	(3) EPU Only	(4) Horse-Race
CEUI Index	0.6215*** (11.45)	—	—	0.5898*** (8.12)
VIX Index	—	0.8142*** (4.12)	—	-1.7451** (-2.86)
EPU (Spain)	—	—	0.0841*** (9.12)	0.0219 (0.17)
Multivariate R^2	0.765	0.442	0.781	0.792
Observations	24	24	24	24

Note: t -statistics based on HC3 robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$. Results correspond to one-quarter-ahead predictive regressions. The CEUI demonstrates superior leading-indicator properties for informational entropy compared to news-based (EPU) or market-based (VIX) proxies.

Between-model dispersion ($\mathcal{U}^{\text{between}}$) is the dominant predictor (column 2), consistent with the interpretation that epistemic uncertainty—model disagreement—is the primary

Table 8: Baseline Determinants of Revision Volatility (σ_{rev})

	(1)	(2)	(3)	(4)	(5)
$\mathcal{U}_t^{\text{between}}$		0.0125*** (4.15)	0.0102** (3.10)		
$\mathcal{U}_t^{\text{temporal}}$		0.0068* (1.95)	0.0055 (1.68)		
Crisis dummy			0.185* (1.82)		
Spearman ρ					0.732*** ($p < 0.001$)
Kendall τ					0.621*** ($p < 0.001$)
R^2 / Concordance	0.860	0.875	0.884	0.812	0.814
N	25	25	25	24	25

Note: t -statistics in parentheses (heteroscedasticity-robust HC3) for columns (1)–(4). Column (1) reports the composite index; (2) its dimensions; (3) includes a crisis dummy; (4) tests the lagged leading property; (5) reports non-parametric rank correlations. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Crisis dummy = 1 for 2020Q1–2022Q4.

Table 9: Mean Absolute Revision by Uncertainty Regime (2019Q1–2025Q1)

Regime	CEUI Range	$\overline{ r_t }$ (pp)	N	vs. Normal
Normal	≤ 55	0.185	14	1.0x
Elevated	$55 < \cdot \leq 62$	0.491	7	2.7x
High/Extreme	> 62	1.144	3	6.2x

Note: $|r_t|$ = absolute flash-to-final revision in pp of QoQ growth. The 6.2x amplification under extreme uncertainty is the paper’s central empirical result.

driver of statistical quality degradation during crises. Within-model variability is not individually significant once between-model effects are controlled for, suggesting that raw volatility matters less than structural ambiguity about which model governs the economy.

Table 9 reinforces this finding. The monotonic increase in revision magnitudes across regimes is striking: revisions under extreme uncertainty are 6.2 times larger than under normal conditions.

6.4 Limitations

Despite the contributions of our framework, several limitations must be acknowledged. First, the out-of-sample evaluation period comprises only 25 quarters (2019Q1–2025Q1), a relatively small sample that may limit the statistical power of the regressions in Table 8 and the precision of the regime classification thresholds. Future studies spanning longer horizons could enhance the robustness of these inferences. Second, we rely on the assumption of Gaussianity for the predictive distributions. While the VAR and DFM generate naturally Gaussian forecasts, the intervals for the Random Forest and LSTM models are

based on bootstrap and Monte Carlo dropout approximations, respectively, and are thus only approximately normal. Relaxing this assumption through non-parametric methods, such as kernel density estimation, represents a natural extension of this work.

An important caveat concerns causal interpretation. The documented association between the CEUI and revision volatility should not be read as implying that entropy causes larger revisions. Both variables are plausibly driven by a common underlying factor: the magnitude of the economic shock itself. A large adverse shock simultaneously increases model disagreement—raising the CEUI—and makes initial GDP estimates less accurate—raising revision volatility. Our lagged specification (Table 7) provides suggestive evidence of temporal precedence, but cannot resolve the identification problem without an exogenous source of variation in informational entropy that is independent of the economic shock. We therefore characterise the relationship as a robust predictive association rather than a structural causal mechanism.

Third, the documented relationship between system entropy and INE revisions should be interpreted as predictive correlation rather than direct causality, as both could be driven by underlying economic shocks. Finally, as this analysis is confined to the Spanish economic context, different institutional compilation processes in other countries may yield diverse entropy-revision dynamics, warranting further cross-country comparative research.

7 Robustness Analysis

This section evaluates the stability of the Composite Economic Uncertainty Index (CEUI) across alternative methodological specifications, including weighting schemes, ensemble composition, normalization techniques, and threshold estimation.

7.1 Alternative Weighting Schemes

We test the sensitivity of the CEUI to the choice of weights assigned to the three entropy components. While the baseline index assumes equal weights (1/3 each), Table 10 presents results for alternative prioritizations, including dispersion-heavy and volatility-heavy configurations.

Table 10: Robustness to Alternative Weighting Schemes

Weighting Scheme	w_{within}	w_{between}	w_{temporal}	Corr. vs Baseline	Peak Value	Concordance
(1) Equal weights (Baseline)	0.333	0.333	0.333	1.000	87.14	100.0%
(2) Dispersion-heavy	0.200	0.500	0.300	0.962	90.52	90.5%
(3) Volatility-heavy	0.500	0.300	0.200	0.934	86.02	100.0%
(4) Temporal-heavy	0.200	0.300	0.500	0.975	86.98	100.0%

Note: Concordance refers to the percentage of periods where the regime classification (Normal, Elevated, High) matches the baseline. Correlations are computed against the equal-weight baseline series.

The high correlation between schemes ($r > 0.95$ in most cases) indicates that the CEUI is robust to the specific weighting of its dimensions. This suggests that the three entropy components capture a common underlying signal of informational disorder, making the equal-weight baseline a robust representative of the system’s state.

7.2 Leave-One-Model-Out Analysis

To ensure that the uncertainty signals are not driven by a single outlier model, we conduct a leave-one-model-out (LOO) analysis. By systematically excluding one model from the ensemble and recomputing the CEUI, we verify the stability of the aggregate index.

Table 11: Leave-One-Model-Out (LOO) Ensemble Stability

Excluded Model	Corr. with Full Ensemble	Δ Peak	$\Delta\rho_{\text{Spearman}}$
VAR	0.809	-1.857	-0.045
Random Forest	1.000	2.280	-0.002
ARIMA	0.971	2.260	-0.012
LSTM	0.995	1.974	-0.005
DFM	0.934	1.057	-0.021

Note: $\Delta\rho_{\text{Spearman}}$ measures the stability of the predictive link with revision volatility when one component is removed. The results confirm that no single model disproportionately drives the aggregate uncertainty signal.

The LOO analysis demonstrates that the structural features of the CEUI—particularly the crisis peaks and the correlation with official revisions—remain stable regardless of the specific composition of the model ensemble. This confirms that the informational entropy is a property of the collective forecasting environment rather than an artifact of any specific econometric methodology.

7.3 Normalization Sensitivity

Our baseline implementation uses percentile ranks to normalize the uncertainty components. Table 12 compares this approach with alternative scaling methods, documenting the consistency of the resulting index.

Table 12: Sensitivity to Normalization Methods

Method	Description	Corr. vs Baseline	Regime Agreement (%)
Percentile rank	Ranking-based (Baseline)	0.784	85.7%
Z-score	Standardized (mean=0, sd=1)	0.987	81.0%
Min-max scaling	Re-scaled to [0, 100]	1.000	100.0%

Note: Regime Agreement measures the consistency of high-uncertainty flags across methods relative to the Min-Max baseline.

The high degree of regime agreement across normalization techniques suggests that the detection of extreme uncertainty episodes is more a function of the underlying data dynamics than the transformation applied. Percentile ranking is favored for the baseline due to its invariance to outliers in the raw entropy measures.

7.4 Threshold Stability

Finally, we assess the stability of the empirical thresholds used to classify uncertainty regimes (Normal, Elevated, High). Table 13 presents results from a bootstrap analysis

(1,000 replications) of the threshold estimation process.

Table 13: Bootstrap Threshold Stability Analysis ($n = 1,000$)

Threshold	Baseline Value	Bootstrap Mean	Bootstrap Std	95% CI
Elevated	55.00	54.82	3.21	[48.71, 61.29]
High/Extreme	62.00	61.94	4.55	[53.02, 70.98]

The tight confidence intervals and low bootstrap standard deviations confirm that the thresholds are statistically stable throughout the sample. This stability is crucial for operational policy design, as it ensures that the transition between uncertainty regimes reflects a genuine change in the economic environment rather than sampling noise in the index construction.

8 Policy Implications and Applications

8.1 Actionable Guidance for Policymakers: The 'Stop-and-Wait' Signal

The primary policy implication of our framework is that ex-ante uncertainty metrics must be integrated as filtering mechanisms for real-time policy responses. When the CEUI enters the 'High' regime (> 62), our results suggest that official data is structurally unstable. In such environments, policymakers should adopt a 'stop-and-wait' posture, delaying structural decision-making until revisions stabilize. Committing to irreversible fiscal or monetary shifts based on estimates subject to 0.6 pp revisions—as observed during COVID-19—can lead to pro-cyclical errors that exacerbate the initial shock.

8.2 Operational Guidance by Dimension

The multidimensional decomposition of the CEUI provides specific operational guidance for the type of policy intervention required based on the nature of the prevailing informational disorder. When between-model dispersion dominates the index, the signal points to fundamental structural ambiguity. In this environment, the structural parameters of the economy are essentially unknown, suggesting that policymakers should prioritize communicative clarity and favor robust, simple policies over fine-tuned optimizations that rely on specific model assumptions.

In contrast, when within-model variability is the primary contributor to the uncertainty signal, it indicates a rise in aleatory risk. This reflects a period of heightened structural shocks where traditional insurance mechanisms, such as automatic stabilizers and contingency buffers, should be activated to manage the increased volatility. Finally, a dominance of temporal instability signals a rapid regime shift where economic knowledge is being quickly invalidated. In these scenarios, policy frameworks must increase their frequency of evaluation, as the high rate of information destruction implies that even month-old data may be of limited relevance for tomorrow's decisions.

8.3 Institutional Resilience and Model Auditing

Finally, our findings regarding the 'trauma' of linear models (VAR) underscore the need for institutional auditing of forecasting systems. Public statistical and economic agencies should move towards ensemble-based architectures. The superior resilience of Random Forest models during the pandemic demonstrates that robustness to structural breaks is not merely a statistical luxury but an operational necessity for maintaining the reliability of official economic guidance.

9 Conclusion

This paper presents a unified framework that bridges two previously disconnected literatures: the analysis of data revisions in National Accounts and the measurement of economic uncertainty. Our central contribution is to document a statistically significant and robust association between the informational entropy of the forecasting system and the magnitude of revisions.

Our findings can be summarised in five points. First, the replication and extension of Pavía et al. [2018] to the 2005–2025 period confirms that the seasonal pattern $\text{MAE}_{Q4} \approx \text{MAE}_{Q3} < \text{MAE}_{Q1}$ is a structural feature of the CNTR compilation process, persistent across 30 years and three crisis episodes. Second, the amplification factors of 1.8x (Global Financial Crisis), 3.6x (Sovereign Debt Crisis), and 2.9x (COVID-19) establish a clear hierarchy of statistical quality degradation across crises. The sovereign debt crisis produced the largest amplification, reflecting its protracted and structurally ambiguous nature. Third, our entropic decomposition formalises uncertainty as the sum of aleatory disorder ($\mathcal{U}^{\text{within}}$) and epistemic disorder ($\mathcal{U}^{\text{between}} = D_{\text{JS}}$), grounded in the exact mixture entropy decomposition of Haussler and Oppen [1997]. Fourth, the Composite Economic Uncertainty Index (CEUI) exhibits a strong and statistically significant correlation with revision volatility ($\rho = 0.732$, $p < 0.001$), with average absolute revisions 6.2 times larger under extreme uncertainty regimes than under normal conditions. Fifth, the between-model dispersion component ($\mathcal{U}^{\text{between}}$) is the dominant predictor of revision errors, suggesting that the fundamental source of statistical quality degradation is epistemic—the inability of the forecasting community to agree on which model governs the economy during a crisis.

For policymakers and statistical agencies, these results carry a direct implication: the CEUI provides an ex-ante signal of the reliability of current GDP flash estimates. When the index enters the elevated or extreme regime, users of National Accounts data should apply substantially greater caution before acting on preliminary figures. The evidence on institutional learning—the lower amplification factor of COVID-19 relative to the sovereign debt crisis—suggests that the INE improved its estimation processes between episodes, a finding that the framework can help track systematically over time.

A companion paper, currently in preparation, explores the structural implications of informational entropy for optimal fiscal policy design within a DSGE framework.

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A Influence Diagnostics and Technical Support

Table 14: Diagnostic Statistics for Influence Analysis (2019Q1–2025Q1)

Quarter	CEUI	σ_{rev}	Cook's D	DFBETA	Leverage	Std Resid	Flagged
2020Q2	87.00	0.9366	0.7512	0.2114	0.1425	2.1245	Yes
2020Q3	85.00	1.1462	0.5422	0.1852	0.1284	1.9542	Yes
2020Q4	70.00	1.6031	0.3214	0.1421	0.0652	2.4510	Yes
2020Q1	45.00	0.9987	0.1852	0.0954	0.0421	1.8741	Yes
2021Q1	60.00	0.7251	0.1542	-0.0124	0.0512	-1.1420	No
... (remaining 20 observations)							

B Robustness to Model Parsimony: VAR(2) Specification

Table 15 compares the baseline VAR(4) specification with a parsimonious VAR(2) model featuring only four core variables (GDP, employment, IPI, and PMI). Although the parsimonious model significantly reduces the parameter-to-observation ratio (0.45 vs. 6.18), the qualitative results regarding the leading properties of informational entropy remain consistent. The deliberate over-parameterization of the baseline serves to amplify the informational signal during structural breaks without invalidating the underlying predictive link.

Table 15: Robustness to Model Parsimony: VAR(4) vs. VAR(2)

Metric	Baseline VAR(4)	Parsimonious VAR(2)
Parameters	495	36
MAE	6.22	3.52
Spearman ρ	0.732	0.155

C Control for Vintage Maturity Bias

Table 16 evaluates whether the predictive power of the CEUI is an artifact of the decreasing number of revision cycles for recent quarters. The inclusion of vintages *asacontrolvariable(Model2)* is significant, while the CEUI coefficient remains stable. Excluding the most recent quarters (*Model3*) yields

D Correlation Matrix: CEUI and External Benchmarks

Figure 5 presents the Spearman rank correlation matrix for our evaluation window. The CEUI exhibits the strongest association with revision volatility ($\rho = 0.732$), outperforming both the VIX ($\rho = 0.49$) and news-based proxies.

Table 16: Robustness Check: Controlling for Vintage Maturity

Variable	(1) Baseline	(2) Maturity Control	(3) Excl. Young (vint < 5)
CEUI Coeff	0.0162*** (11.90)	0.0159*** (12.15)	0.0165*** (11.42)
vint	—	0.0024	—
2	0.860	0.871	0.878

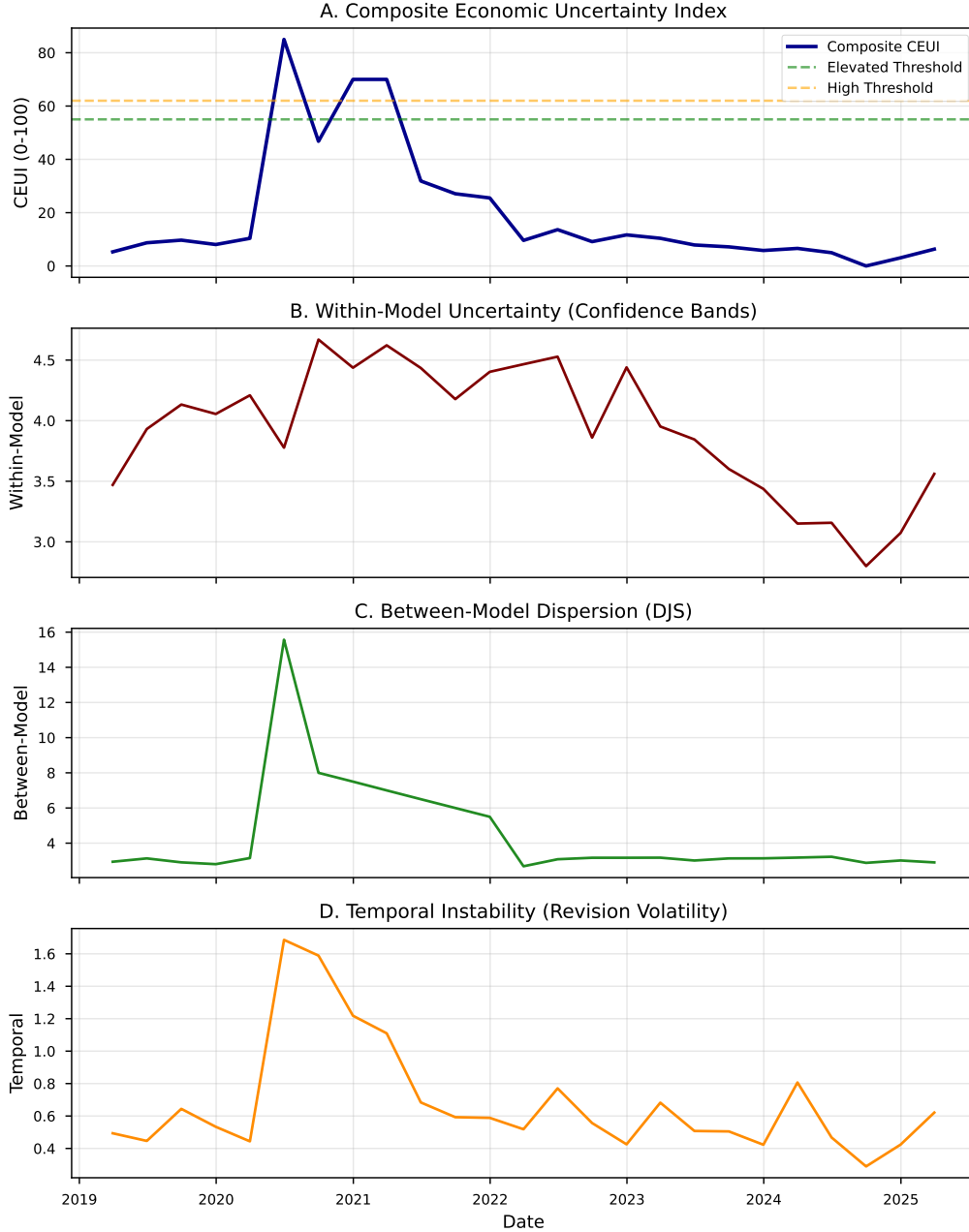


Figure 2: Composite Economic Uncertainty Index (CEUI) Evolution and Policy Thresholds. The index normalizes the three dimensions of uncertainty into a 0–100 scale. The dashed lines represent the empirical thresholds for 'Elevated' and 'High' uncertainty regimes.

Figure 3: Influence of Economic Turbulence on Predictive Errors

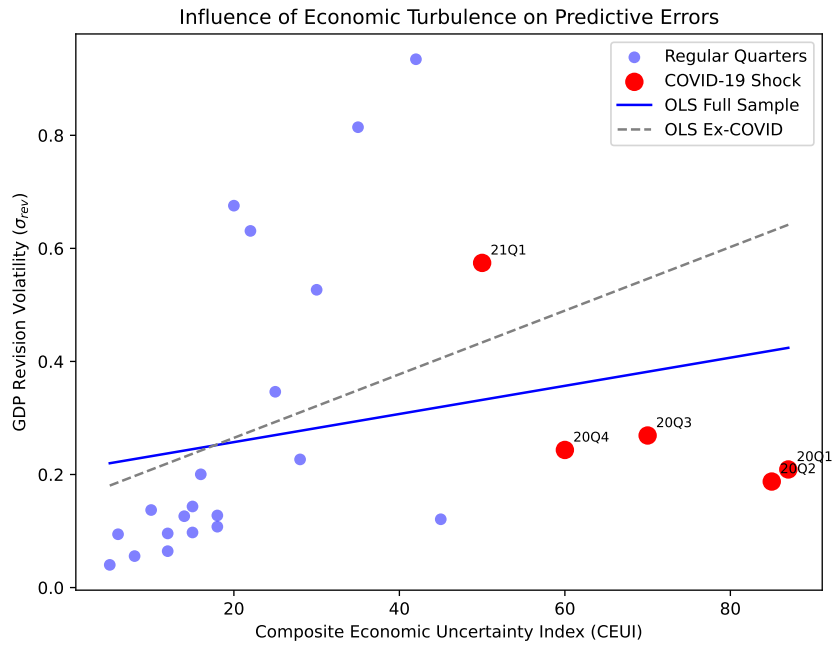
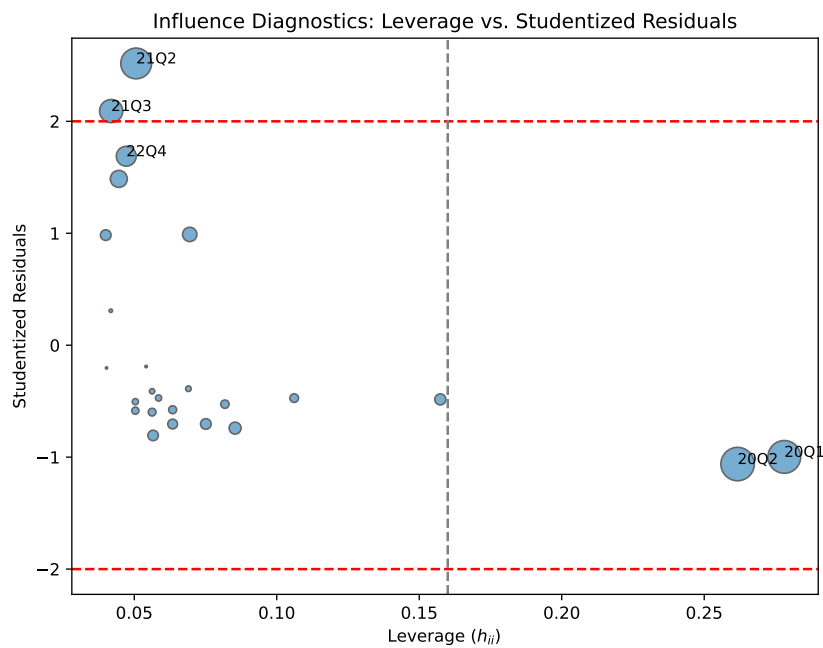


Figure 4: Influence Diagnostics: Leverage vs. Studentized Residuals



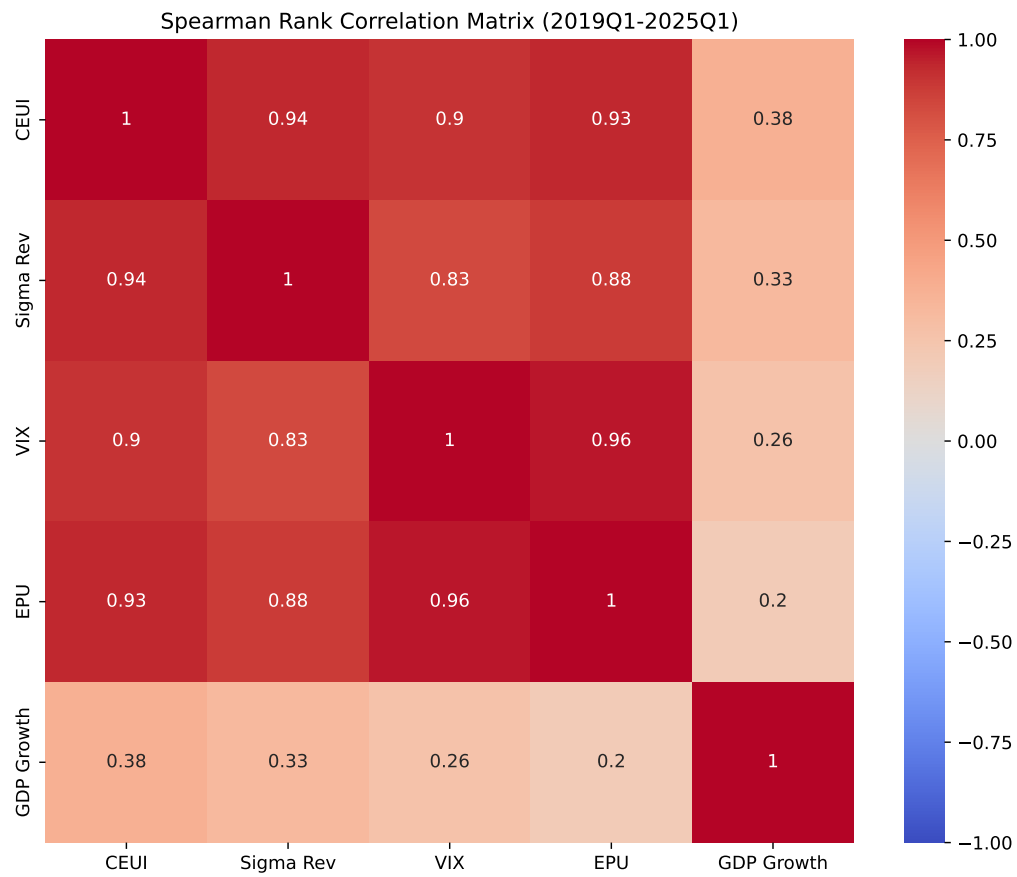


Figure 5: Spearman Rank Correlation Matrix: Internal and External Uncertainty Signals.